

XIII LAW3M - Abstract Submission Template

dipolar compass-needle arrays as a controllable analogue for spin-ice dynamics: hysteresis, damping, and magnetic avalanches

João P. Sinnecker

Centro Brasileiro de Pesquisas Físicas (CBPF), Rio de Janeiro, Brazil

Artificial spin-ice arrays of lithographically patterned magnetic nanoislands provide a direct route to study how local dipolar coupling produces collective magnetic states [1]. However, in nanofabricated systems the effective damping is a material property and cannot be tuned independently of geometry, spacing, and switching-barrier disorder. One may ask, therefore, why not simulate monodomain nanoisland arrays directly. The answer is that a macroscopic compass needle obeying Newton's second law for rotation, $I\ddot{\theta}_i = \tau_i^{\text{dip}} + \tau_i^{\text{ext}} - b\dot{\theta}_i$, provides something nanoisland models cannot: a single, continuously tunable control parameter — the quality factor $Q = \omega_0 I/b$, where $\omega_0 = \sqrt{mB_{\text{ref}}/I}$ is the natural precession frequency set by nearest-neighbour dipolar coupling — that allows the same dipolar lattice to be driven from the overdamped limit ($Q \ll 1$, slow monotonic reversal, narrow loops) to the underdamped limit ($Q \gg 1$, inertial overshoot and chain-reaction reversal cascades), and directly through the critical-damping crossover ($Q \approx 1$) by changing only b . This systematic tunability is what motivates the use of compass-needle arrays [3] as a minimal, transparent model for spin-ice dynamics. Here we use it to address a question that remains open in the artificial-spin-ice literature [2]: does the dynamical regime set by damping control the universality class of magnetic avalanches, and does the answer depend on the lattice geometry? Simulations are performed on 30×30 square, triangular, and honeycomb arrays — macroscopic analogues of the square, kagome, and brick-wall spin-ice geometries — sweeping Q over two decades and applying five-ramp hysteresis cycles. Avalanche size distributions are extracted and fitted by maximum-likelihood power-law estimation, and waiting-time distributions between successive events are analysed to discriminate between Poisson and scale-free (SOC-like) inter-event timing. Preliminary results already reveal the central signal: the power-law exponent τ increases systematically with Q in all three geometries, from $\tau \approx 1.2$ – 1.3 near the crossover to $\tau \approx 1.7$ – 1.8 in the underdamped limit, while the hysteresis loop area and coercive field grow monotonically as the system becomes overdamped. The triangular lattice additionally shows non-monotonic domain-size behaviour with Q absent from the other geometries, a candidate signature of frustration-inertia coupling. These results suggest that inertial dynamics, commonly neglected in spin-ice models, is a relevant variable for classifying the avalanche universality of dipolar arrays, and the compass-needle platform provides a clean experimental route to test this in the laboratory.

[1] S. H. Skjærvø *et al.*, Nat. Rev. Phys. **2**, 13 (2020).

[2] K. A. Dahmen, Y. Ben-Zion and J. T. Uhl, Nat. Phys. **7**, 554 (2011).

[3] R. L. Novak and J. P. Sinnecker, J. Magn. Magn. Mater. **272–276**, 1557 (2004).